

Imperial College London
Department of Earth Science and Engineering
MSc in Environmental Data Science and Machine Learning

Independent Research Project
Project Plan

Multi-Entity Visual Grounding for Complex Remote Sensing Scenes

by
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repository: <https://github.com/ese-ada-lovelace-2025/irp-sy1124.git>

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Abstract

Multi-entity Remote Sensing Visual Grounding (RSVG) under complex instructions frequently causes Vision Language Models (VLMs) to suffer from attention drift and "spatial hallucinations." Existing works have attempted to use reinforcement learning to optimize VLMs, but remain limited by sparse, outcome-oriented rewards, which inadvertently induce image-detached pseudo-reasoning and reward hacking. To address these critical bottlenecks, this project proposes an end-to-end grounding framework based on a Single-Pass Visual Tree-of-Thought (Visual ToT) and Process Reward Models (PRM).

This research constructs tree-structured spatial search priors via reverse trajectory synthesis. During the reinforcement learning phase, multidimensional visual process rewards are introduced to provide dense supervision for intermediate reasoning behaviors. Ultimately, this approach aims to dismantle the black-box blind guessing mechanisms of conventional VLMs and establish a highly interpretable, process-oriented target grounding framework. By simulating genuine human visual search behaviors, this method is expected to significantly improve multi-entity localization accuracy in complex overhead scenarios.

1 Introduction

Remote sensing foundation models are evolving from basic pixel-level classification to high-level semantic reasoning [Chen et al., 2025]. However, recent benchmarks like OmniEarth [Zhang et al., 2026] and GEOBench-VLM [Smith and Doe, 2024] highlight a critical defect: existing VLMs are highly prone to "spatial hallucinations" when processing complex spatial topologies and multi-entity long-text instructions.

Currently, supervised fine-tuned (SFT) VLMs rely on implicit "black-box decoding," directly predicting coordinates without transparent physical verification. While recent studies introduce Group Relative Policy Optimization (GRPO) to address this, they still face three fatal limitations:

First, Task Formulation Bias. Existing prompt templates strictly preset reference objects in spatial order [Zhang et al., 2026], which artificially plans the search path and severely disconnects from unstructured, real-world queries. Second, Representation Disconnect. Forcing models to separately output lengthy `<think>` and `<answer>` blocks exacerbates the "lost in the middle" phenomenon [Liu et al., 2024]. Purely textual "spatial derivation" fails to form genuine bidirectional constraints with the visual encoder, devolving into a performance that merely caters to templates. Third, Reward Misalignment. Existing GRPO methods rely solely on the final Intersection over Union (IoU) as a sparse, outcome-oriented reward. As demonstrated by Lightman et al. [2023], this lacks fine-grained gradient guidance for multi-step visual reasoning, easily inducing "reward hacking" [Skalse et al., 2022].

To resolve these bottlenecks, this project proposes an end-to-end framework. Inspired by explicit reasoning research [Brown and Miller, 2025], this method discards black-box decoding and explicitly constructs a visually-integrated search tree comprising four stages: Semantic Planning, Candidate Generation, Self-Reflection, and Target Grounding. By continuously verifying visual results during the search, it actively focuses spatial attention, mitigating attention drift and tightly binding the reasoning process to the final grounding results.

Furthermore, we employ reverse trajectory synthesis to construct tree-structured spatial prior data [Johnson and Lee, 2025]. During the GRPO phase, we replace the single IoU reward with a multidimensional visual PRM engine [Wu and Zhou, 2025], providing dense, step-level constraint signals. Ultimately, this establishes an interpretable, process-oriented framework

poised to advance visual grounding in complex scenarios.

2 Methods

2.1 Dataset Selection and Prior Trajectory Synthesis

High-quality data with complex spatial topologies is a prerequisite for training a model’s logical reasoning capabilities [Gütter et al., 2022]. This project utilizes remote sensing datasets with graded difficulty levels: DIOR-RSVG for basic image-text perception, and multi-entity nested datasets like OmniEarth to stimulate logical potential.

To generate the requisite ”intermediate reasoning labels” absent in existing datasets, we design a reverse trajectory synthesis pipeline [Garcia and Martinez, 2025] (detailed in Algorithm 1). Specifically, large language models are utilized to reversely deconstruct instructions into macro, meso, and micro hierarchical levels. Combined with open-vocabulary detectors like Grounding DINO [Liu et al., 2023] to extract candidate bounding boxes, geometric scripts are employed to weave these into structured thought trajectory data, providing high-quality priors for supervised fine-tuning.

2.2 Single-Pass Visual Tree-of-Thought Inference Architecture

Building upon the need to mitigate attention drift in long contexts [Liu et al., 2024], the proposed autoregressive Single-Pass Visual ToT architecture forces the model to complete explicit reasoning within a single forward pass [Brown and Miller, 2025]. The workflow is executed sequentially: performing logical dimensionality reduction at the text level, conducting a breadth-first scan to generate candidate regions, combining visual features to logically verify and prune erroneous boxes, and finally outputting precise coordinates.

To formally define this mechanism, we formulate it as a Markov Decision Process (MDP). Given an input image I and a complex query Q , the generated thought trajectory is $\tau = \{s_0, a_1, s_1, a_2, \dots, a_T, s_T\}$. The joint probability of generating the correct reasoning path is:

$$P(\tau \mid I, Q) = \prod_{t=1}^T \pi_{\theta}(a_t \mid s_{t-1}, I, Q) \quad (1)$$

This ensures that each reasoning action a_t is strictly conditioned on the historical reasoning state s_{t-1} , structurally preventing spatial hallucinations (see Figure 1).

Algorithm 1 Phase 1: Reverse Trajectory Synthesis & Supervised Fine-Tuning

Input: Raw image-text pairs $\mathcal{D} = \{(I, Q, B_{gt})\}$, Open-Vocabulary Detector $\mathcal{O}$, LLM $\mathcal{M}$

Output: SFT-initialized VLM $\pi_{\theta_{SFT}}$

- 1: Initialize synthesized dataset $\mathcal{D}_{SFT} \leftarrow \emptyset$
 - 2: **for each** $(I, Q, B_{gt}) \in \mathcal{D}$ **do**
 - 3: $P \leftarrow \mathcal{M}(Q)$ {Deconstruct into semantic plan: $L_{macro}, L_{meso}, L_{micro}$ }
 - 4: $C_{pool} \leftarrow \mathcal{O}(I, P)$
 - 5: $\tau_{verify} \leftarrow \{\text{Verify}(c, B_{gt}) \rightarrow [\text{Proceed}] \text{ or } [\text{Pruned}] \mid c \in C_{pool}\}$
 - 6: $\tau_{SFT} \leftarrow P \oplus C_{pool} \oplus \tau_{verify} \oplus B_{gt}$
 - 7: $\mathcal{D}_{SFT} \leftarrow \mathcal{D}_{SFT} \cup \{(I, Q, \tau_{SFT})\}$
 - 8: **end for**
 - 9: **return** $\pi_{\theta_{SFT}}$
-

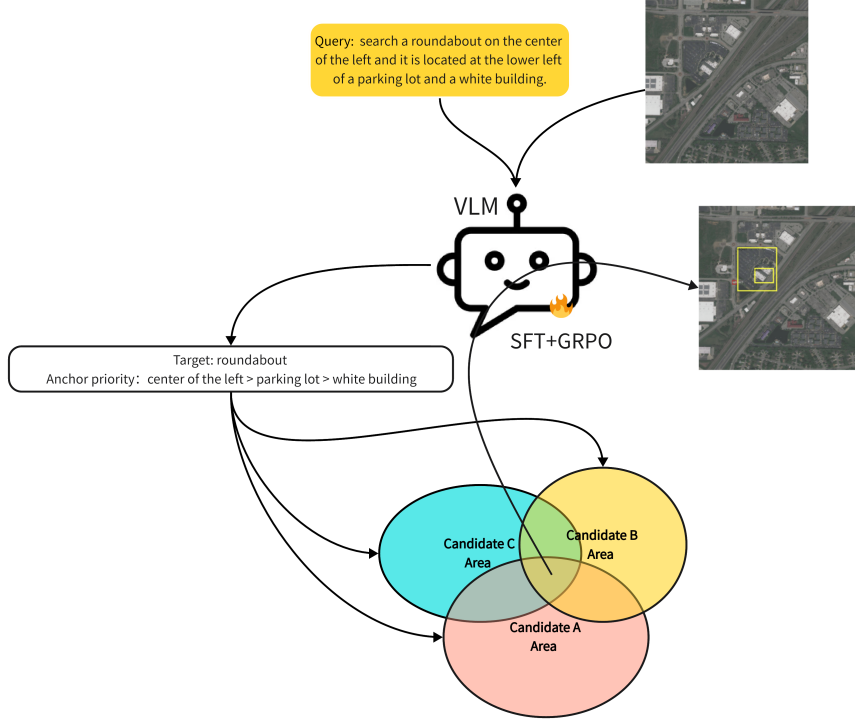


Figure 1: Overview of the Visual Tree-of-Thought architecture for complex spatial reasoning.

2.3 Reinforcement Learning Alignment Based on Multi-Level Process Rewards

To overcome the limitations of sparse outcome rewards and reward hacking [Skalse et al., 2022], this project introduces a Hierarchical Process Reward Model (PRM) mechanism [Lightman et al., 2023]. The optimization objective is refined to provide dense supervision for each decision step (outlined in Algorithm 2). In addition to evaluating grounding accuracy, the reward engine emphatically incorporates a "planning reward," a "recall reward," and crucially, a "pruning reward" when the model successfully uses logic to eliminate erroneous distractors.

For any sampled trajectory τ , the total reward R_{total} is defined as the weighted sum of these process rewards:

$$R_{total} = \lambda_1 R_{Plan} + \lambda_2 R_{Recall} + \lambda_3 R_{Pruning} + \lambda_4 R_{Final-GIoU} \quad (2)$$

This dense formulation directly guides the model to master rigorous spatial-temporal deductive logic.

2.4 Comprehensive Evaluation Metrics

Unlike previous studies that solely focused on final accuracy [Nipko and Davis, 2020], this project establishes a comprehensive evaluation system emphasizing both outcomes and processes. Process verification metrics such as "Path Success Rate" are introduced to objectively examine whether the model genuinely eliminated distracting entities during reflection.

Furthermore, to provide continuous gradients for geometric precision, we employ the Generalized Intersection over Union (GIoU). Let P be the predicted bounding box, G be the ground

Algorithm 2 Phase 2: PRM-Guided GRPO Alignment

Input: SFT-initialized model $\pi_{\theta_{SFT}}$, Training queries $\mathcal{D}_{RL} = \{(I, Q, B_{gt})\}$ **Output:** Optimized Visual ToT Model π_{θ^*}

```
1: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_{SFT}}$  and reference policy  $\pi_{ref} \leftarrow \pi_{\theta_{SFT}}$ 
2: while not converged do
3:   Sample a batch of inputs  $(I, Q, B_{gt}) \sim \mathcal{D}_{RL}$ 
4:   Sample  $G$  ToT trajectories  $\{\tau_i\}_{i=1}^G \sim \pi_{\theta}(\cdot | I, Q)$ 
5:   for  $i = 1$  to  $G$  do
6:     Parse  $\tau_i$  into sub-components: Plan  $P_i$ , Candidates  $C_i$ , Verification  $V_i$ , Final Box  $B_i$ 
7:      $R_{tot}^i \leftarrow \lambda_1 R_{Plan}(P_i) + \lambda_2 R_{Recall}(C_i) + \lambda_3 R_{Prune}(V_i) + \lambda_4 R_{GIoU}(B_i)$ 
8:   end for
9:   Compute group relative advantages  $A_i = (R_{tot}^i - \mu_R) / \sigma_R$  for  $i = 1 \dots G$ 
10:  Compute probability ratio  $\rho_i = \frac{\pi_{\theta}(\tau_i | I, Q)}{\pi_{ref}(\tau_i | I, Q)}$ 
11:  Update  $\theta$  to maximize the GRPO objective:
12:     $\mathcal{J}_{GRPO}(\theta) = \mathbb{E} \left[ \sum_{i=1}^G \min(\rho_i A_i, \text{clip}(\rho_i, 1 - \epsilon, 1 + \epsilon) A_i) - \beta \mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) \right]$ 
13: end while
14: return  $\pi_{\theta^*}$ 
```

truth, and C be the smallest enclosing convex hull. GIoU is formulated as:

$$GIoU(P, G) = \frac{|P \cap G|}{|P \cup G|} - \frac{|C \setminus (P \cup G)|}{|C|} \quad (3)$$

2.5 Implementation Details and Baselines

To ensure computational feasibility within the project timeframe, we will adopt open-weight VLMs, such as Qwen3-VL or InternVL, as the base policy models. Parameter-Efficient Fine-Tuning (PEFT) techniques, specifically Low-Rank Adaptation (LoRA), will be strictly utilized to mitigate the heavy GPU memory overhead associated with multi-stage RL training. The training pipeline will be implemented utilizing the HuggingFace `trl` library and deployed on high-performance computing clusters.

3 Work Plan

	June First half	June Second Half	July First Half	July Second Half	August First Half	just Second
Phase 1: Preparation & Data Synthesis						
1.1 Architecture & Literature						
1.2 SFT Data Reverse Synthesis						
Phase 2: Model Training & Alignment						
2.1 Supervised Fine-Tuning (SFT)						
2.2 Geometric Judge (PRM) Construction						
2.3 PRM-GRPO Alignment						
Phase 3: Evaluation & Ablation Studies						
3.1 Core Benchmark Testing						
3.2 Ablation Study						
Phase 4: Documentation & Presentation						
4.1 Draft Report Writing						

Figure 2: Project work plan and timetable.

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